

Top-Levi: Multi-User Interactive System Using Acoustic Levitation

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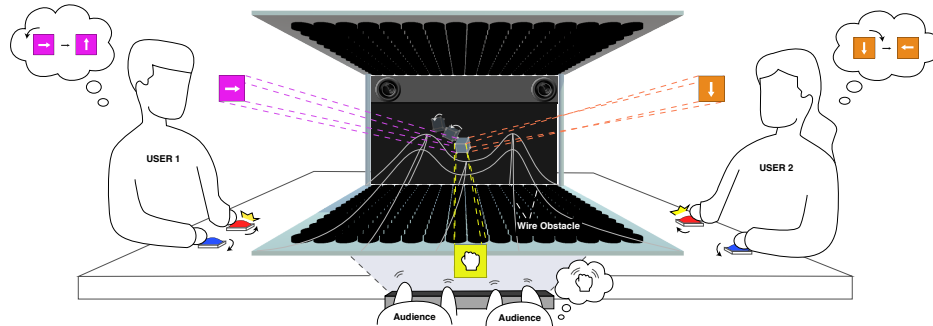


Figure 1: Overview of *Top-Levi*: In *Top-Levi* a floating cube is moved along a curved path. 1) Users on the left and right sides are in charge of steering the cube by tapping buttons in sequences shown on the sides of the cube. 2) The audience is responsible for generating power to move the cube forward by waving their hands in response to specific hand gestures on the side of cube.

ABSTRACT

Top-Levi is a public multi-user interactive system that requires a pair of users to cooperate with an audience around them. Based on acoustic levitation technology, *Top-Levi* leverages a unique attribute of dynamic physical 3D contents displayed and animated in the air: such systems inherently provide different visual information to users depending on where they are around the device. In *Top-Levi*, there are two primary users on opposite (left/right) sides of the device, and audience members to the front. Each sees different instructions displayed on a floating cube. Their collective task is to cooperate to move the cube from a start point to a final destination by synchronizing their responses to the instructions.

KEYWORDS

Acoustic levitation, Physical 3D visualisation, Multi-user interactive system

ACM Reference Format:

Jiwan Kim and Hyunjae Gil. 2022. Top-Levi: Multi-User Interactive System Using Acoustic Levitation. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3526114.3561347>

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UIST '22 Adjunct, October 29–November 2, 2022, Bend, OR, USA

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ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3561347>

1 INTRODUCTION

Acoustic levitation is a novel technology that can display and animate physical objects in mid-air. Early examples of these devices could acoustically trap light objects at specific locations by using the mechanical energy of ultrasound [5, 13]. More advanced ultrasound displays [8, 10] can move various types of object along three-dimensional trajectories, such as polystyrene beads [7–9], fabric [1, 7], or liquid [2, 12]. This is achieved by dynamically changing the distribution of ultrasound forces around the objects in order to achieve the desired transformations [1, 6]. The promising technology enables the display of dynamic free-floating physical 3D visualizations [9, 11] and can be enhanced with interactive graphics by projecting images onto the levitated objects [3, 7]. More recently, researchers used a twin trapping system to realize a volumetric display based on a fast levitating particle, and diffuse illumination [3, 4] and physically animated levitated structures through the use of acoustically transparent fabric strung between beads [1, 7].

The idea of dynamic physical 3D visualizations presents a number of interesting and unique properties. One of these is that displayed content can be seen from any point around the display. As such, what a particular user will see depends on both their field of view on the display and the specific orientation of the object. In this way, a single levitating object can deliver different visual information to users situated at different angles around it. We were inspired by this property to develop *Top-Levi*. *Top-Levi* provides a multi-user interactive game based on navigating a levitating cube along a curved trajectory—steering it through a simple maze. It involves two users, who control travel direction, and an audience that provides power. The users and audience each occupy a different position around the cube, thus seeing different information shown on its sides. Based on what they see, they need to cooperate with each other to move the cube along the path to its destination.

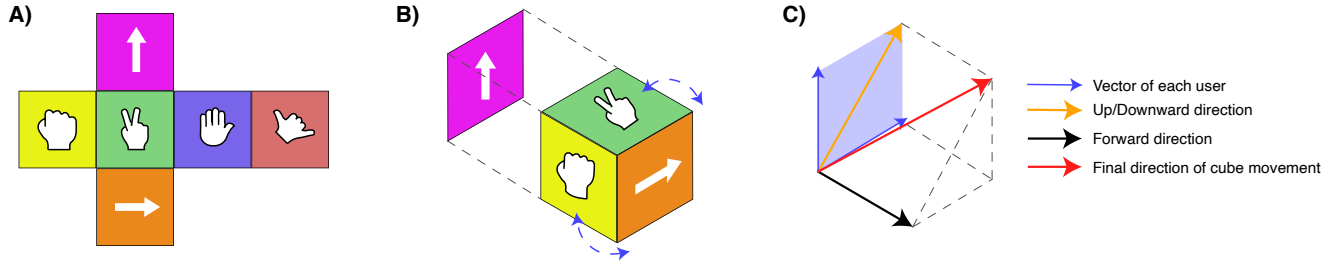


Figure 2: Design of *Top-Levi Plane*. (A) *Top-Levi* shows different visual information painted on each side. (B) It rotates by two main users' cooperation, and (C) moves along the final direction (red arrow) by the audience's mass interaction.

2 TOP-LEVI

We propose a multi-user interactive system using the unique attributes of a physical 3D display (Figure 1). Our system requires the cooperation of users based on the different visual information they see on the sides of a levitating cube. The main goal is to collaboratively move the cube along a curved path to its destination. Figure 2 illustrates the design of the cube and explains how it moves.

Two main users are in charge of steering the cube. Each user sees the current direction they can move the cube in by an arrow painted on the side. The arrows are not visible to the other user or audience and are orthogonal to one another. The overall movement of the cube is the sum of the two arrow vectors (Figure 2-C). To navigate the cube along its path (shown by thin wire obstacles), the two users need to rotate the cube to the appropriate direction (Figure 1). They achieve this simply by tapping physical buttons mounted in front of them: respectively red and blue buttons to rotate the cube clockwise and counter-clockwise in the plane parallel to the users.

Meanwhile, the audience is responsible for generating the power required to move the cube along the vector specified by the two users (Figure 2-C). Each of the four cube faces that can be presented to the audience shows a different gesture: an open palm, a scissor, a fist, and a telephone. The audience is required to wave their hands with a specific gesture they currently see (Figure 1). Since the cube rotates according to the input of the two main users, the audience is required to keep changing their gestures as the orientation of the cube changes. A hand tracking system detects the number of waving hands in the audience, which determines the speed at which to move the cube.

In summary, our idea includes the following interactive features:

- A levitating cube that delivers different visual information to two users and the audience via physically manipulating a static display (e.g., rotating a cube with different images painting/printing on its sides)
- A physical user interface for two users to cooperatively manipulate the movement direction of the levitating cube.
- A gesture input interface for an audience to drive the movement of the levitating cube.

3 IMPLEMENTATION

Top-levi uses an Ultrasound Levitation Kit with its API and UNITY framework provided by UIST Student Innovation Contest. We levitate and manipulate the cube structure with a multi-point levitation algorithm suggested by Morales et al. [7]. The cube consists of six acoustically transparent fabrics and used eight polystyrene beads as anchors. The particle separation is 30mm to achieve a physical display surface that will support clearly visible printed material. The fabric is kept rigid by maintaining high stiffness between the beads. Different visual information is painted on each side of the cube. By utilizing a real-time tracking system, we track the bead locations and recognize which side of the cube is displayed to each user. We also set up the curved path for the moving cube to travel along with thin wires which minimally occlude the system's ultrasound waves.

For the user interfaces, we built physical devices composed of a pair of large buttons for the two main users. A microprocessor communicating with the acoustic levitation system captured tapping inputs on these buttons and provided visual feedback via an inbuilt LED. To track the audience, a hand tracking device was placed in front of the display to detect the number and form of the hand gestures made by the audience. To achieve multi-hand tracking, we use a single RGB camera and analyze the hand image in real-time using the hand landmark model in the MediaPipe framework [14].

4 CONCLUSION

Top-Levi is designed around one of the unique characteristics of displaying dynamic physical 3D content in the air: it can provide different visual information to users at different locations. We explore the impact of this property by designing a game in which a pair of users and the audience watching them need to collaborate to steer an object through a simple maze based on the different information shown to them on its sides. We hope our system expands the dialogue about the potential of acoustic levitation technology to achieve new forms of an interactive experience for users.

ACKNOWLEDGMENTS

Thanks to the UIST student innovation contest. This work was supported by Korea Institute for Advancement of Technology (KIAT) grant funded by the Korea Government (MOTIE) (P0012725, The Competency Development Program for Industry Specialist).

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